

Computer Vision (EE6102)

Point Processing

2026 Fall

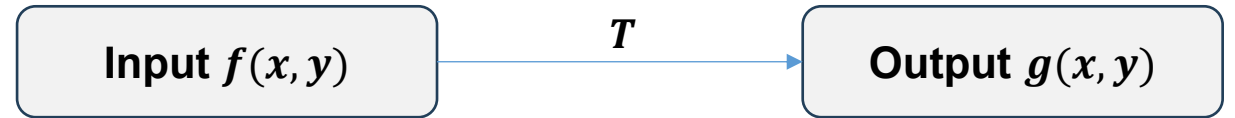
Image Enhancement

- Objective of enhancement
 - Make an image more suitable than the original for a specific application
 - Often for human perception
- Two broad categories
 - Spatial domain
 - Point processing (intensity transforms)
 - Spatial filters
 - (pixel neighborhoods)
 - Frequency domain
 - Working with frequencies

Spatial Domain

$$g(x, y) = T[f(x, y)]$$

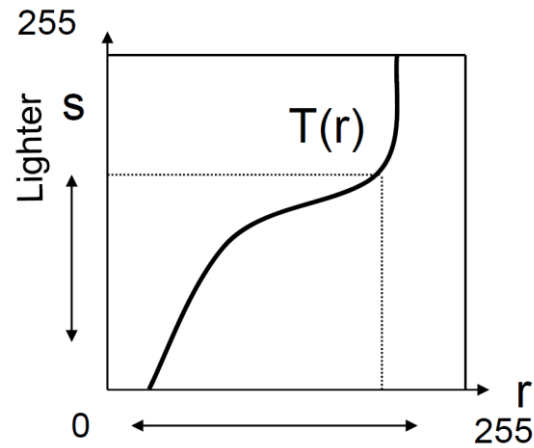
- $f(x, y)$: input image
- $g(x, y)$: output image
- T : operator on f
 - Defined over some *neighborhood* of (x, y)
 - can operate on a set of Images
- $g(x, y) = T\{f(s, t) | s, t \in N(x, y)\}$



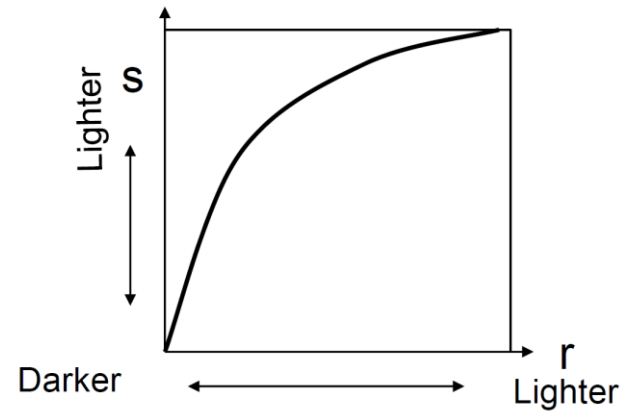
Gray-level Transformation

- Simplest form of T
 - Pixel neighborhood is 1×1
 - When the neighborhood size is 1×1 , the transformation depends only on the intensity of the current pixel. This is called a gray-level transformation or point processing.
- Notation: $s = T(r)$
 - r, s denote gray level of $f(x, y)$ and $g(x, y)$ for any point (x, y)
- Plot of $s = T(r)$
 - T can be
 - **negative, log transform, power-law transform, contrast stretching**

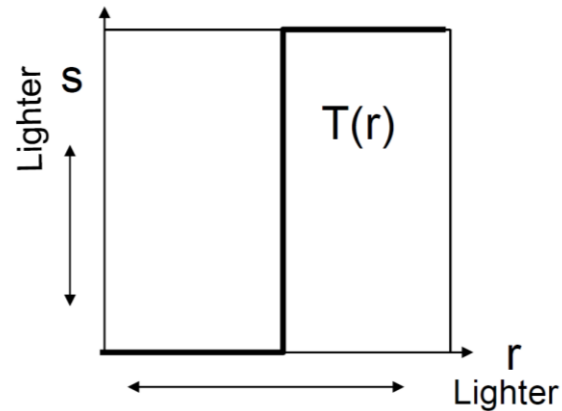
Plot of $s = T(r)$



Input



$$s \approx c \log(1 + r)$$



$$s \approx \begin{cases} 0, & r \leq r_0 \\ 255, & r \geq r_0 \end{cases}$$



Intensity Scaling



$f(x, y)$



$a \cdot f(x, y)$

Scaling in the γ -domain is equivalent to scaling in the linear luminance domain

$$I \sim (af(x, y))^\gamma = a^\gamma (f(x, y))^\gamma$$

Same effect as adjusting camera exposure time

Intensity scaling simply multiplies every pixel by a constant. It changes the overall brightness of the image, similar to changing the camera exposure.

Intensity Scaling

- Gamma response

- The relationship between pixel intensity and displayed luminance can be modeled as

$$I \propto f^\gamma$$

- I : displayed luminance
 - γ : nonlinear response parameter

- After intensity scaling

$$I' \propto (af)^\gamma = a^\gamma f^\gamma$$

- Therefore

$$I' = a^\gamma I$$

Scaling all pixel values by a constant results in a proportional scaling of displayed luminance, even under a nonlinear gamma response

Adjusting γ (Power-law Transformation)

Original image



$f(x, y)$

γ increased by 50%



$$\underbrace{a}_{\text{overall scale}} \cdot \underbrace{(f(x, y))^r}_{\text{nonlinear tone mapping}} \quad \text{with } \gamma = 1.5$$

Same effect as using a different photographic film

- $\gamma = 1$: No change
- $\gamma > 1$: image becomes darker
- $\gamma < 1$: image becomes brighter

Adjusting γ (Power-law Transformation)

$$\underbrace{a}_{\text{overall scale}} \cdot \underbrace{(f(x, y))^{\gamma}}_{\text{nonlinear tone mapping}} \quad \text{with } \gamma = 1.5$$

- $\gamma = 1$: No change
- $\gamma > 1$: Image becomes darker
- $\gamma < 1$: Image becomes brighter



Original Image



$\gamma = 0.5$



$\gamma = 1.0$



$\gamma = 2.0$

Log Transformation

- Expands small intensity differences in dark regions
- Compresses the wide dynamic range in bright regions
- Effective for enhancing shadow details

$$s = c \log(1 + r)$$

r : input pixel intensity

s : output pixel intensity

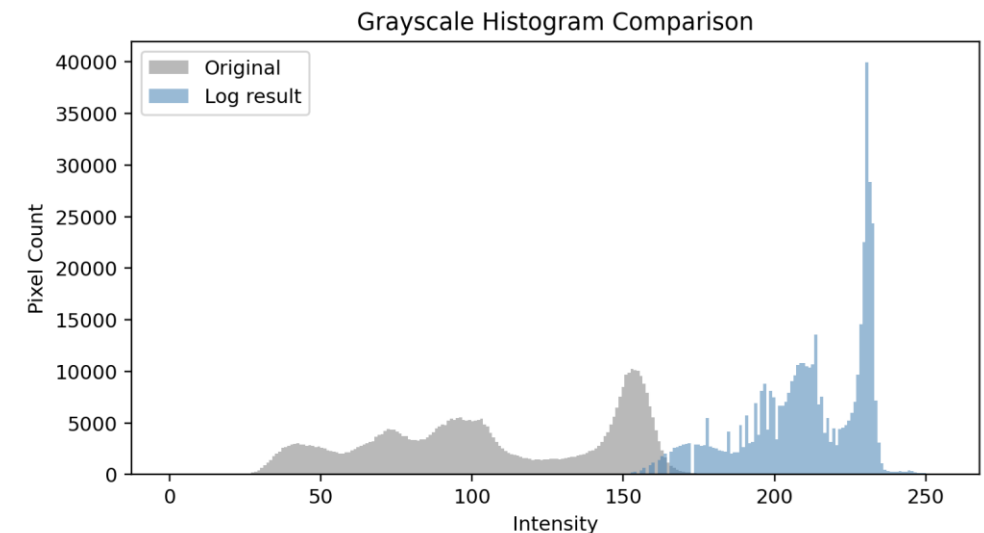
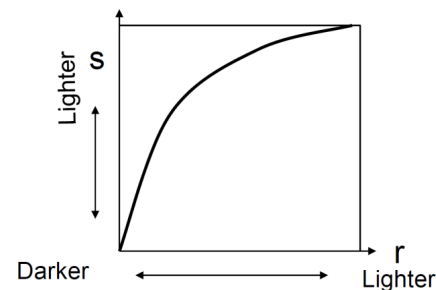
c : scaling constant ($=255/\log(1 + 255)$)



Original Image



Log Transformed Image



Bit-plane Slicing

- **Upper bits:** Major structures and large intensity variations
- **Lower bits:** Fine details and noise-like variations

8-bit grayscale image

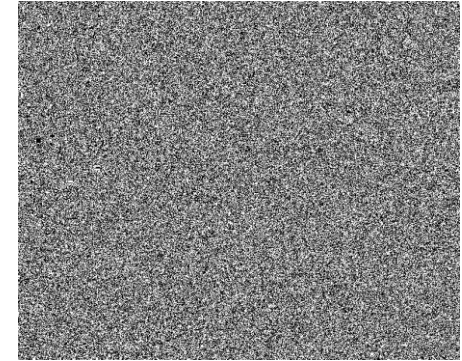
$$I(x, y) = b_7 2^7 + \dots + b_1 2^1 + b_0 2^0$$
$$= \sum_{k=0}^7 b_k(x, y) 2^k \quad (b_k(x, y) \in \{0, 1\})$$

$$b_k(x, y) = \left(\frac{I(x, y)}{2^k} \right) \bmod 2$$

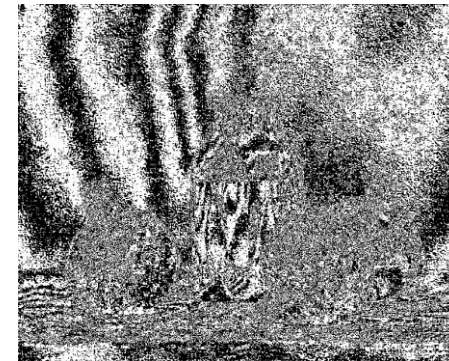
Extracts the k -th bit from the pixel value.



Original Image



$k = 0$ Image



$k = 3$ Image



$k = 6$ Image

Bit-plane Slicing

- $I(x, y) = 173$

- Applications

- **Information Analysis:** Upper bits contain major image structures and significant intensity information, while lower bits often contain fine details and noise-like variations.
- **Compression:** Lower bit-planes can be discarded while preserving much of the overall image structure.
- **Noise Analysis/Removal:** Noise-like variations are often more noticeable in the lower bit-planes.
- **Watermarking / Steganography:** Information can be embedded in the least significant bits (LSBs).
- **Feature Extraction:** Specific bit-planes can be used to emphasize particular structural patterns in an image

Contrast Stretching

Transfer function

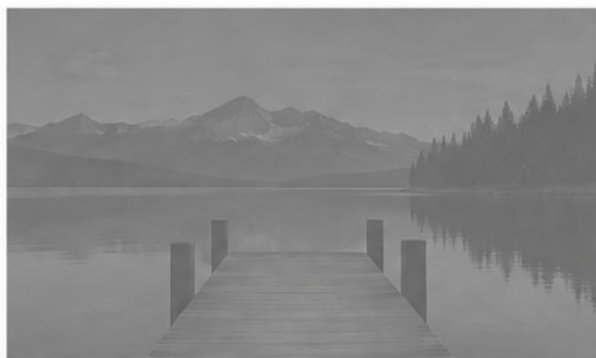
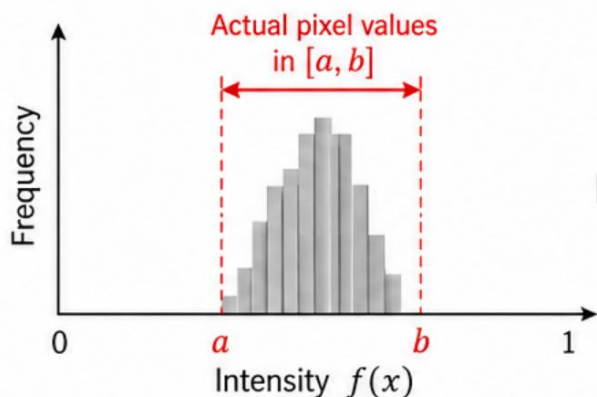
$$g = \phi(f) = \frac{f - a}{b - a}$$

$$a = f_{\min}, \quad b = f_{\max}$$

- Maps the input intensity range $[a, b]$ to the full range $[0, 1]$.
- Linear mapping (point operator).
- $f = a \rightarrow g = 0$
- $f = b \rightarrow g = 1$

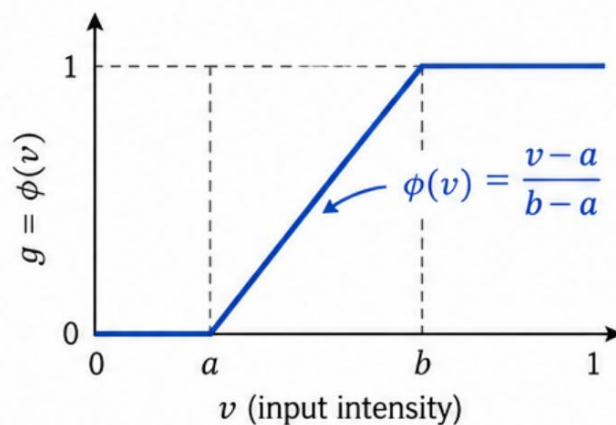
If the original image uses only a limited intensity range $[a, b] \subset [0, 1]$, contrast stretching linearly expands it to the full range $[0, 1]$.

Input intensity histogram (low contrast)

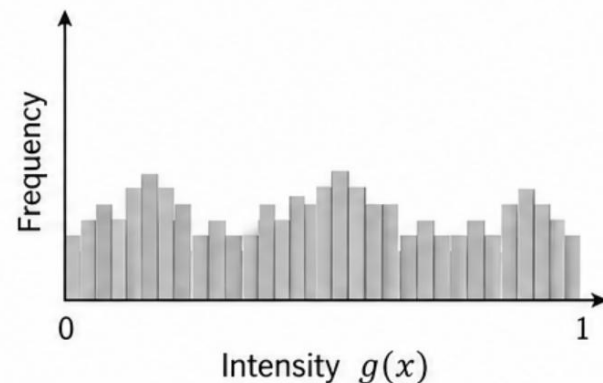


Input image (low contrast)
(pixel values in $[a, b]$)

Transfer function $\phi(v)$



Output intensity histogram (stretched)



Output image (after contrast stretching)
(pixel values in $[0, 1]$)

Overall mapping

$$g(x) = \frac{f(x) - f_{\min}}{f_{\max} - f_{\min}}$$

$$\text{maps } [f_{\min}, f_{\max}] \rightarrow [0, 1]$$

Multiple Noisy Image Averaging

Average model

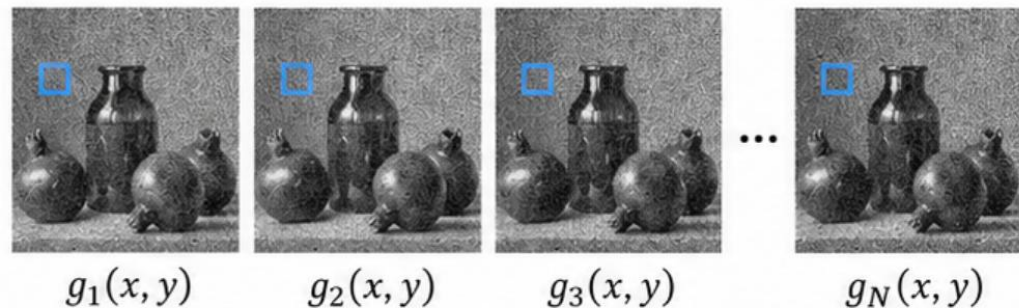
$$\bar{g}(x, y) = \frac{1}{N} \sum_{i=1}^N g_i(x, y)$$

with $g_i(x, y) = f(x, y) + n_i(x, y)$, $\sigma_{avg} = \sigma / \sqrt{N}$

- Average the same pixel location (x,y) across N aligned images
- Independent, zero-mean noise tends to cancel out
- Signal structure is preserved when the images are registered

- If N aligned observations contain independent zero-mean Gaussian noise, average reduces the sd by $1/\sqrt{N}$ while preserving the underlying image signal

Aligned noisy observations



Point-wise average at (x, y)

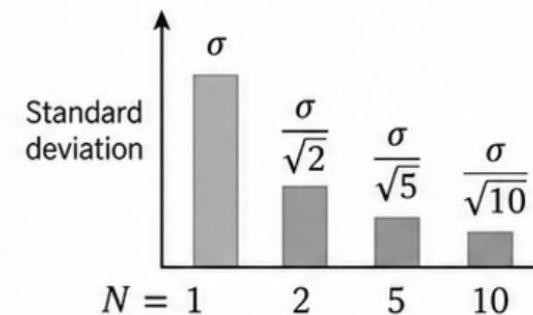
$$\frac{1}{N} \sum_{i=1}^N g_i(x, y)$$

No spatial neighborhood

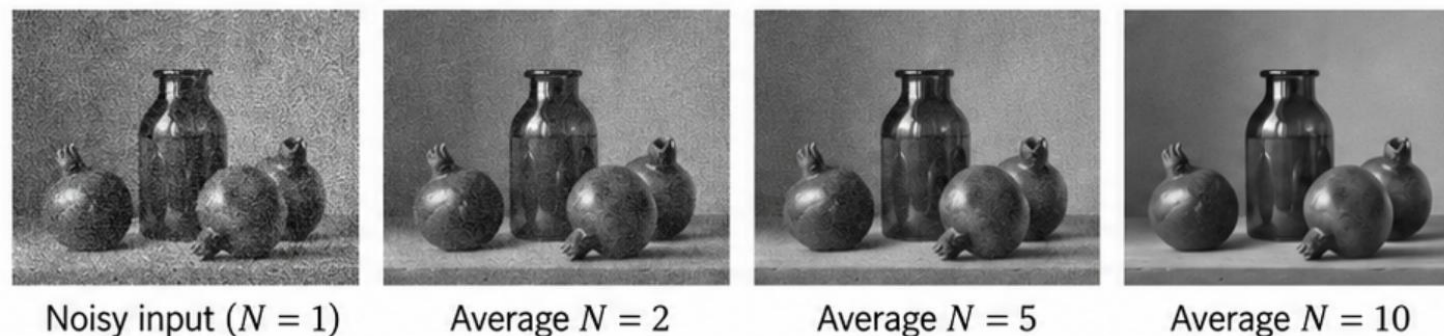
Averaged image



Noise standard deviation



Example: averaging multiple noisy images



Point-wise across images, not neighborhood filtering

$\bar{g}(x, y)$ uses $g_1(x, y), \dots, g_N(x, y)$,
not neighboring pixels such as $g(x \pm 1, y)$.

Multiple Noisy Image Averaging

Assume each noisy observation is

$$g_i(x, y) = f(x, y) + n_i(x, y)$$

where $f(x, y)$ is the true image and $n_i(x, y)$ is independent zero-mean noise:

$$E[n_i(x, y)] = 0, \quad \text{Var}[n_i(x, y)] = \sigma^2$$

The average of N noisy images is

$$\bar{g}(x, y) = \frac{1}{N} \sum_{i=1}^N g_i(x, y)$$

Substituting $g_i(x, y) = f(x, y) + n_i(x, y)$,

$$\bar{g}(x, y) = f(x, y) + \frac{1}{N} \sum_{i=1}^N n_i(x, y)$$

Taking the expectation,

$$E[\bar{g}(x, y)] = f(x, y) + \frac{1}{N} \sum_{i=1}^N E[n_i(x, y)]$$

$$\boxed{E[\bar{g}(x, y)] = f(x, y)}$$

Thus, the signal is preserved on average.

For the noise variance,

$$\text{Var} \left[\frac{1}{N} \sum_{i=1}^N n_i \right] = \frac{1}{N^2} \sum_{i=1}^N \text{Var}[n_i]$$

Since the noise terms are independent,

$$= \frac{1}{N^2} \cdot N\sigma^2 = \frac{\sigma^2}{N}$$

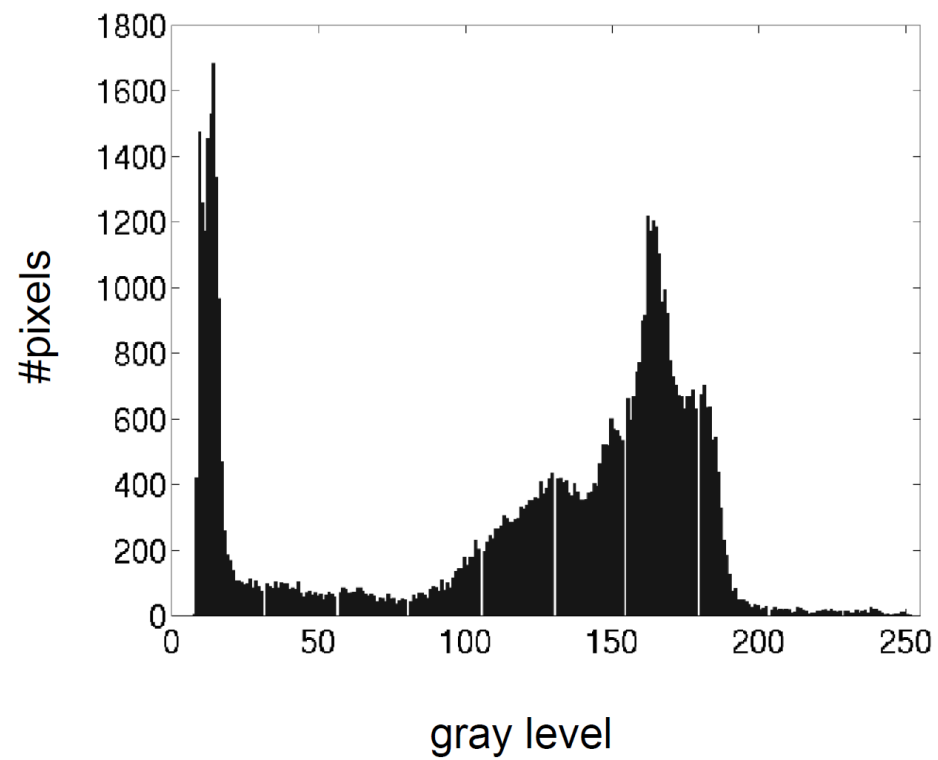
Therefore, the standard deviation of the averaged noise is

$$\boxed{\sigma_{\text{avg}} = \frac{\sigma}{\sqrt{N}}}$$

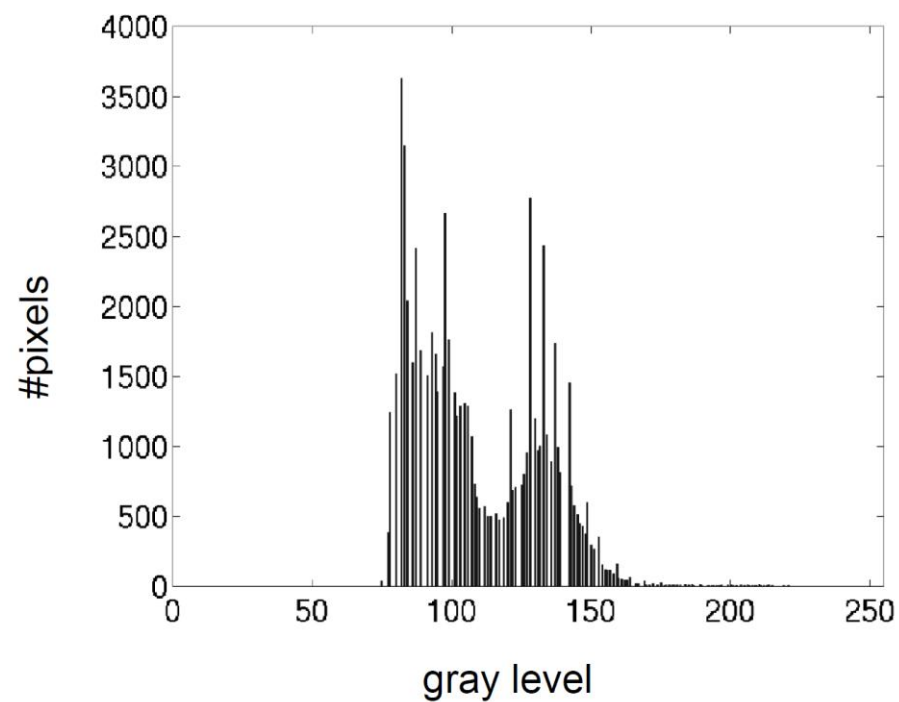
Conclusion: Averaging N aligned noisy images preserves the common signal while reducing independent noise by a factor of $1/\sqrt{N}$.

Histogram

- Distribution of gray-levels can be judged by measuring a histogram:
 - For B-bit image, initialize 2^B counters with 0
 - Loop over all pixels x, y
 - When encountering gray level $f(x, y) = i$, increment counter
 - Histogram can be interpreted as an estimate of the probability density function (pdf) of an underlying random process.
- You can also use fewer, larger bins to trade off amplitude resolution against sample size.



Cameraman
image



Pout
image

Histogram Equalization for Continuous Case

- Idea: find a non-linear transformation

$$g = T(f)$$

to be applied to each pixel of the input image $f(x, y)$, such that a **uniform distribution** of gray level in the entire range results for the output image $g(x, y)$

- Analyze ideal, continuous case first, assuming
 - $0 \leq f \leq 1, 0 \leq g \leq 1$,
 - $T(f)$ is strictly monotonically increasing (one-to-one mapping), hence, there exist

$$f = T^{-1}(g)$$

Goal: pdf $p_g(g) = \text{const.}$ over the range

Histogram Equalization for Continuous Case

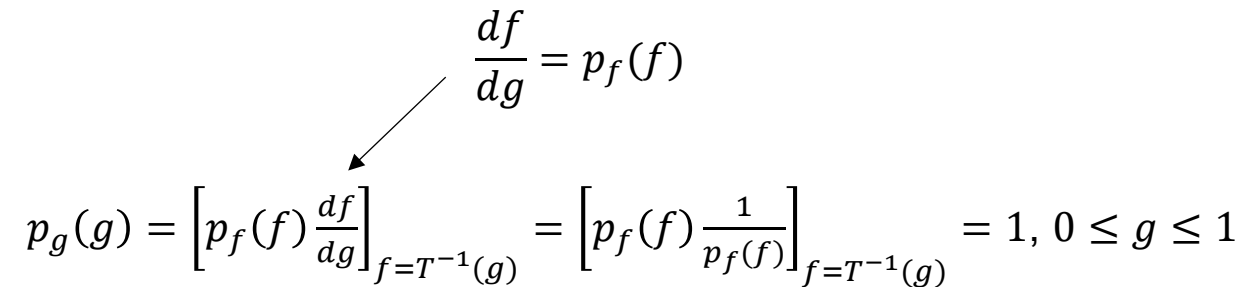
- From basic probability theory

$$p_g(g) = \left[p_f(f) \frac{df}{dg} \right]_{f=T^{-1}(g)}$$

- Consider the transformation function

$$g = T(f) = \int_0^f p_f(\alpha) d\alpha \quad 0 \leq f \leq 1$$

- Then

$$p_g(g) = \left[p_f(f) \frac{df}{dg} \right]_{f=T^{-1}(g)} = \left[p_f(f) \frac{1}{p_f(f)} \right]_{f=T^{-1}(g)} = 1, \quad 0 \leq g \leq 1$$


$$p_g(g) = p_f(f) \frac{df}{dg}$$

$$p_g(g) = 1$$

$$\frac{dg}{df} = p_f(f)$$

$$g = T(f) = \int_0^f p_f(\alpha) d\alpha$$

Histogram Equalization for Discrete Case

- Now, f only assumes discrete amplitude values with $f_0, f_1, \dots, f_{L-1}$ probabilities

$$p_0 = \frac{n_0}{n} \quad p_1 = \frac{n_1}{n} \quad \dots \quad p_{L-1} = \frac{n_{L-1}}{n}$$

- Discrete approximation of $g = T(f) = \int_0^f p_f(\alpha) d\alpha$

$$g_k = T(f_k) = \sum_{i=0}^k P_i$$

- The resulting value are in the range $[0, 1]$ and need to be scaled and rounded appropriately

Histogram Equalization

- Pixel-wise computation of HE
 - For each pixel (x, y) :
 1. Read the input intensity : $r = f(x, y)$
 2. Use the CDF-based transformation corresponding to that intensity

$$T(r) = \sum_{j=0}^r p(r_j)$$

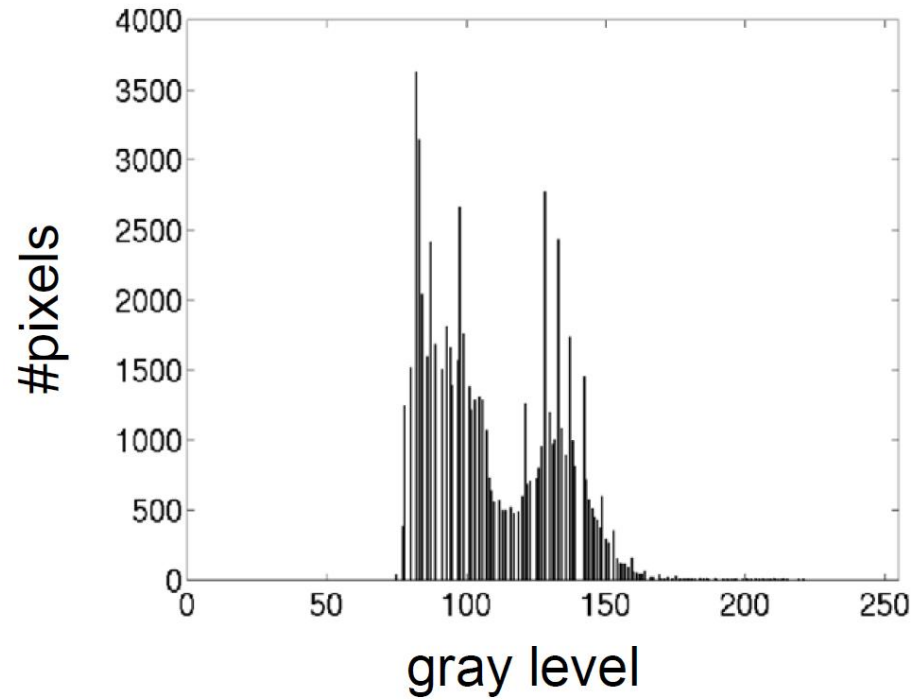
3. Map it to the output intensity range (8-bit grayscale: $L = 255$)

$$g(x, y) = (L - 1)T(f(x, y))$$

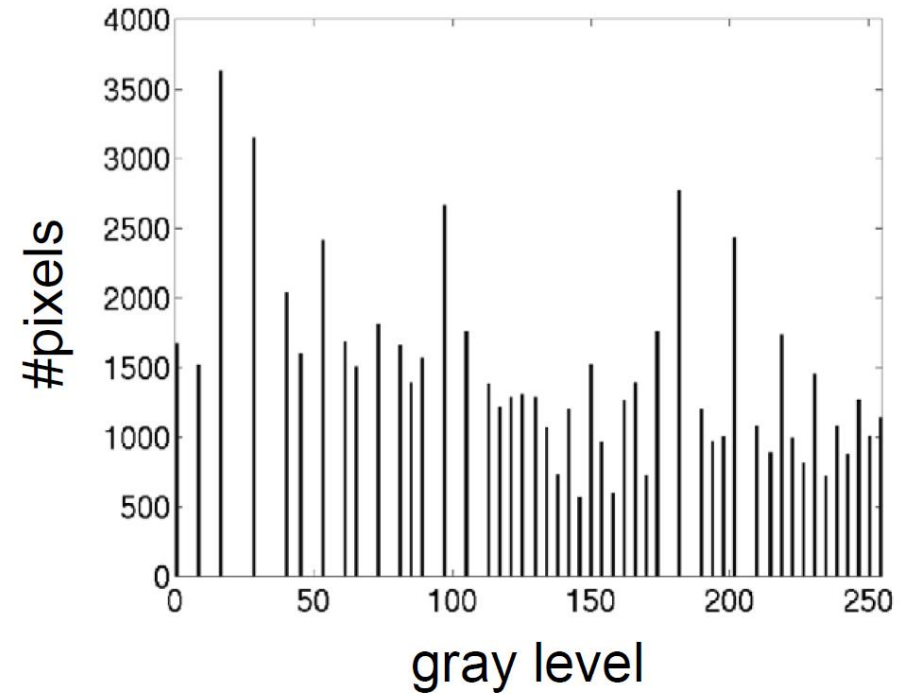
4. For each pixel, use its intensity value as the input to the CDF-based mapping function and replace it with the corresponding output intensity.

Histogram Equalization

Original image *Pout*

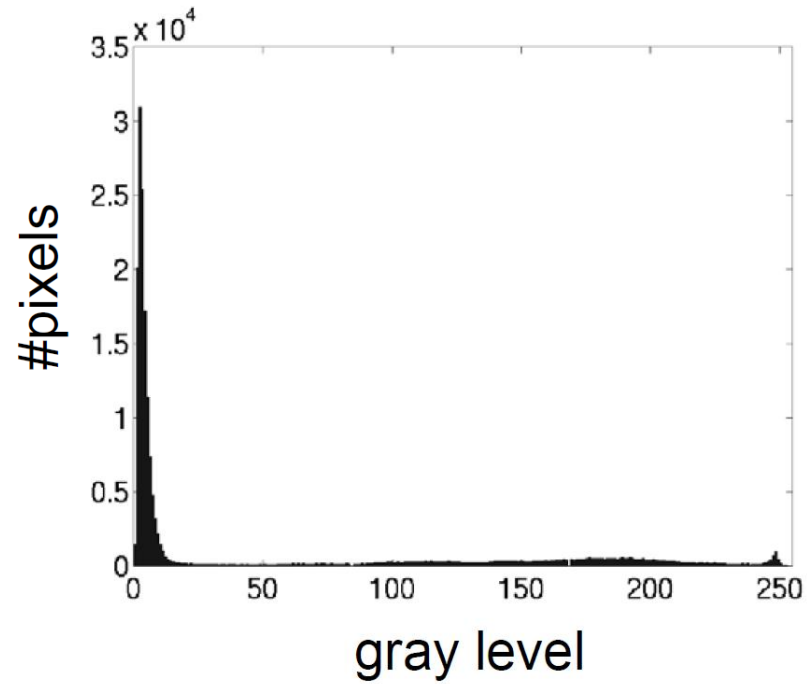


... after histogram equalization

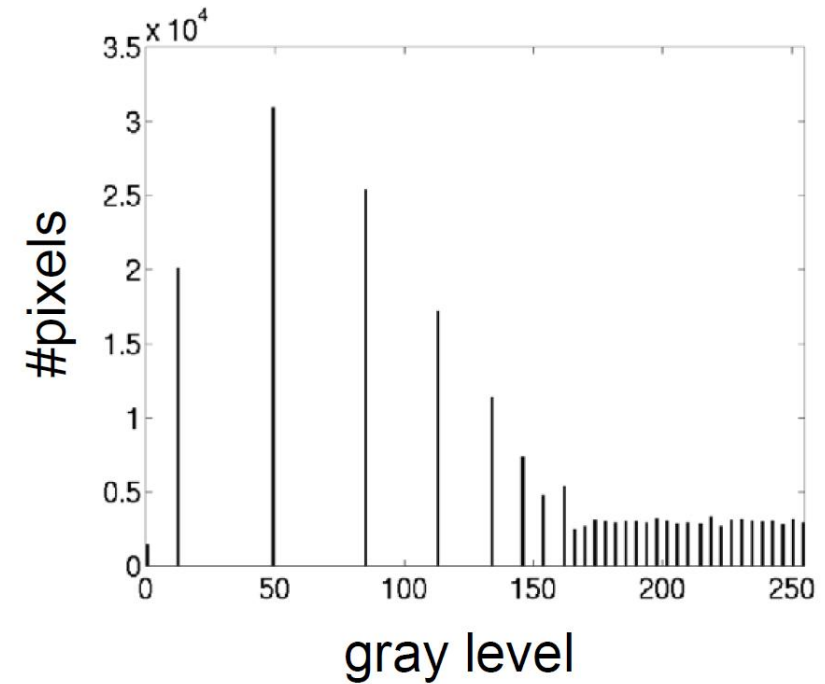


Histogram Equalization

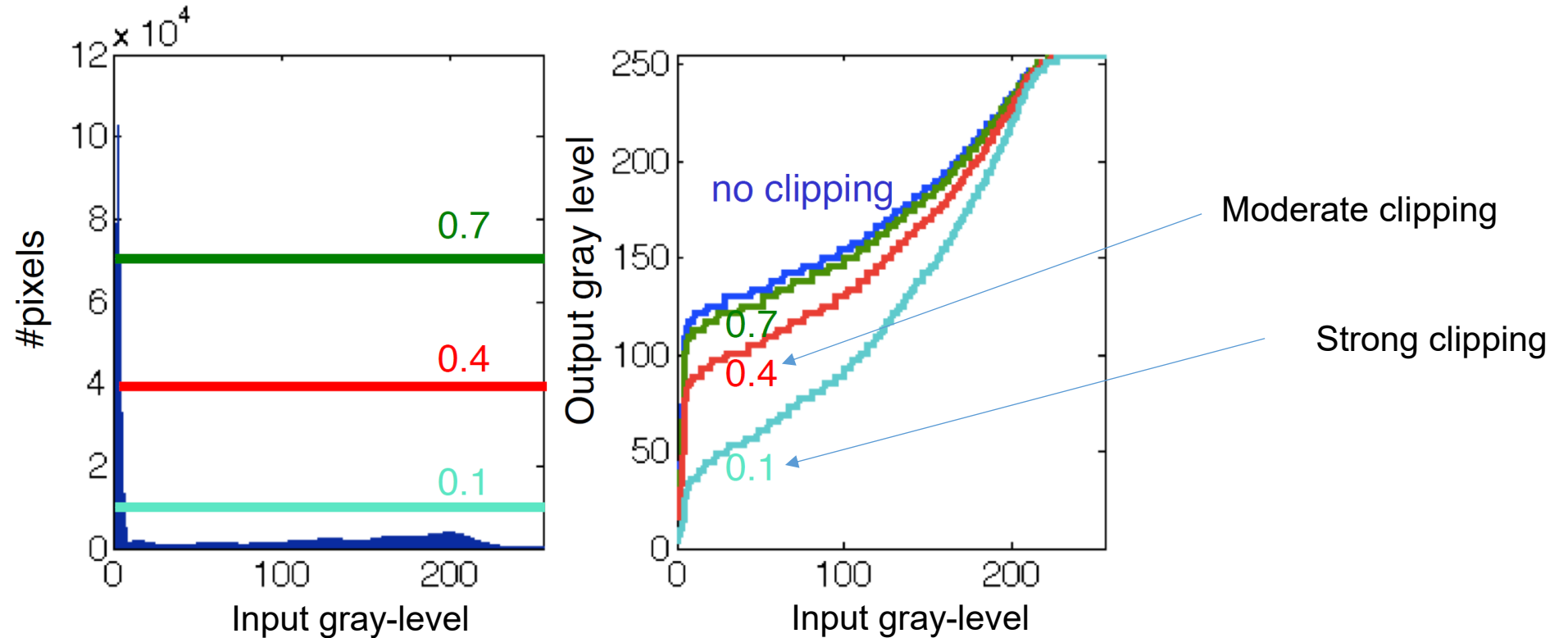
Original



After HE



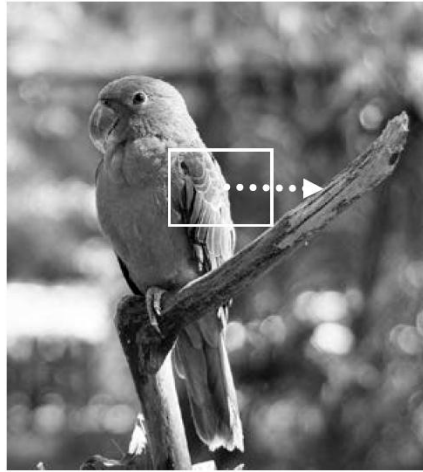
Contrast-Limited Adaptive Histogram Equalization (CLAHE)



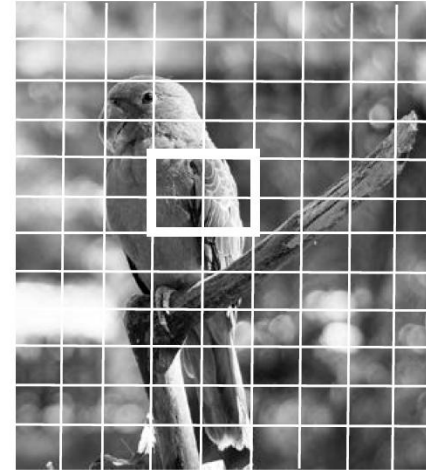
1. **No clipping:** Strong contrast enhancement, but background noise is also significantly amplified.
2. **Moderate clipping:** Provides a good balance between contrast enhancement and noise amplification.
3. **Strong clipping:** Reduces noise amplification, but also weakens contrast enhancement.

Adaptive Histogram Equalization

- Histogram equalization based on a histogram obtained from a portion of the image



Sliding window approach:
different histogram (and
mapping) for every pixel



Tiling approach: subdivide into
overlapping regions, mitigate blocking
effect by smooth blending between
neighboring tiles

- Limit contrast expansion in flat regions of the image, e.g., by clipping histogram values. (“Contrast-limited adaptive histogram equalization”)

CLAHE = AHE + Clipping

Adaptive Histogram Equalization

Original image
Parrot



Global histogram
equalization



Adaptive histogram
equalization, 8x8 tiles



Adaptive histogram
equalization, 16x16 tiles



Original image
Dental Xray



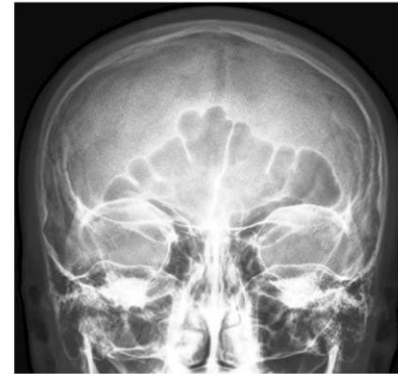
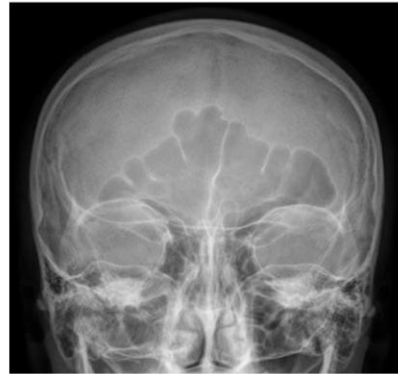
Global histogram
equalization

Adaptive histogram
equalization, 8x8 tiles



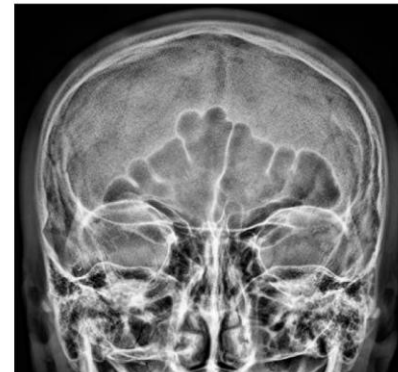
Adaptive histogram
equalization, 16x16 tiles

Original image
Skull Xray



Global histogram
equalization

Adaptive histogram
equalization, 8x8 tiles



Adaptive histogram
equalization, 16x16 tiles